How-To Library

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Configuring IAM Authentication for PowerExchange for Amazon S3

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Abstract

You can use AWS Identity and Access Management (IAM) to control individual and group access to Amazon S3 resources. You can configure AWS IAM to run tasks on the Data Integration Service that is installed on the EC2 system. This article describes the guidelines to configure IAM Authentication for PowerExchange for Amazon S3.

Supported Versions

- PowerExchange® for Amazon S3 10.2

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Overview

To control the access of Amazon S3 resource, you can define permissions to the users by configuring AWS Identity and Access Management (IAM). The AWS IAM service provides enhanced security.

Perform the following steps to configure IAM authentication:

1. Create a minimal Amazon S3 bucket policy.
2. Create the Amazon EC2 Role.
3. Create the Amazon EMR Role.

   Note: To use AWS IAM authentication when you run a mapping in the EMR cluster, you must create the Amazon EMR Role. Then, assign the Amazon ERM Role to the EMR cluster.
4. Create an Amazon S3 connection.

Create a Minimal Amazon S3 Bucket Policy

The minimal Amazon S3 bucket policy restricts user operations and user access to a particular Amazon S3 bucket by assigning an AWS IAM policy to the users. You can configure the AWS IAM policy through the AWS console.

You can use the following minimum required actions when you use PowerExchange for Amazon S3 to successfully read data from and write data to Amazon S3 bucket:

- PutObject
- GetObject
- DeleteObject
- ListBucket
- GetBucketPolicy
- ListAllMyBuckets

The following snippet shows a sample Amazon S3 bucket policy for PowerExchange for Amazon S3:

```
{
"Effect": "Allow", "Action": [ "s3:PutObject", "s3:GetObject", "s3:DeleteObject",
"s3:ListBucket", "s3:GetBucketPolicy" ], "Resource": [ "arn:aws:s3:::<specify_bucket_name>/*",
"arn:aws:s3:::<specify_bucket_name>" ]
}
```

Create the Amazon EC2 Role

You must create an Amazon EC2 Role to provide users access to the Amazon S3 resources. You can use the Amazon EC2 Role when you create an EC2 system in the Amazon S3 bucket.

1. Log in to the **AWS Console**.
2. Click **Dashboard** from the left panel.

   The **AWS Service** dashboard page appears.
3. Click **IAM**.

   The **Welcome to Identity and Access Management** page appears.
4. Click **Policies** from the left panel.

   The **Policy** page appears.
5. Click **Create Policy** or select the required existing Amazon S3 Policy.

   You can edit or review the policy.
6. Select **Role** from the left panel and click **Create role**.

   The **Create role** page appears.
7. Select **EC2** under the **Choose the service that will use this role** section.
8. Select the required **Amazon EC2** role type under the **Select your use case** section.
9. Click **Next: Permission**.
10. Select the required **Amazon S3 Policy** in the **Attach Permission Policies** page.
11. Click **Next: Review**.
12. Specify the name of the role in the **Create role** review page.
13. Click **Create Role**.
14. Review the Role ARN, Instance Profile ARNs, and Policy values in the **Summary** page.

After you create the Amazon EC2 Role, create an EC2 instance. Assign the Amazon EC2 Role to the EC2 instance. For more information about creating an EC2 instance and assigning an Amazon EC2 Role to the Amazon EC2 instance, see the Amazon S3 documentation.

Create the Amazon EMR Role

You can use AWS IAM authentication when you run a mapping in the EMR cluster.

To use AWS IAM authentication in the EMR cluster, you must create the Amazon EMR Role. You must assign the Amazon EMR Role to the EMR cluster for secure access to Amazon S3 resources.

Note: Before you configure IAM Role with EMR cluster, you must install the Informatica Services on an EC2 instance with the IAM Roles assigned.

1. Log in to the **AWS Console**.
2. Click **Dashboard** from the left panel.

   The **AWS Service** dashboard page appears.

3. Click **IAM**.

   The **Welcome to Identity and Access Management** page appears.
4. Click **Policies** from the left panel.

   The **Policy** page appears.
5. Click **Create Policy** or select the required existing Amazon S3 Policy.

   You can edit or review the policy.
6. Select **Role** from the left panel and click **Create role**.

   The **Create role** page appears.
7. Select **EMR** under the **Choose the service that will use this role** section.
8. Select the required Amazon EMR cluster under the **Select your use case** section.
9. Click **Next: Permission**.
10. Select the required **Amazon S3 Policy** in the **Attach Permission Policies** page.
11. Click **Next: Review**.
12. Specify the name of the role in the **Create role** review page.
13. Click **Create Role**.
14. Review the Role ARN, Instance Profile ARNs, and Policy values in the **Summary** page.

Creating an Amazon S3 Connection

When you set up an Amazon S3 connection, you must specify the connection properties for configuring the IAM authentication to control secure access of Amazon S3 resources.

1. In the Developer tool, click **Window** > **Preferences**.
2. Select **Informatica** > **Connections**.
3. Expand the domain in the **Available Connections**.
4. Select the connection type **Enterprise Application** > **Amazon S3**, and click **Add**.
5. Enter a connection name and an optional description.
6. Select Amazon S3 as the connection type.
7. Click **Next**.
8. Configure the following Amazon S3 connection properties in the **Details** tab:

Property	Description
Folder Path	The complete path to Amazon S3 objects. The path must include the bucket name and any folder name. Do not use a slash at the end of the folder path. For example, `<bucket name>/<my folder name>`.
AWS Access Key ID	Not applicable when you use AWS IAM authentication.
AWS Secret Access Key	Not applicable when you use AWS IAM authentication.

Property	Description
Master Symmetric Key	Optional. Provide a 256-bit AES encryption key in the Base64 format when you enable client-side encryption. You can generate a key using a third-party tool.
Customer Master Key ID	Optional. Specify the customer master key ID or alias name generated by AWS Key Management Service (AWS KMS). You must generate the customer master key for the same region where Amazon S3 bucket reside. You can specify either a customer generated customer master key or a default customer master key.
Region Name	Select the AWS region in which the bucket you want to access resides.

9. Click **Test Connection** to verify the connection to Amazon S3.
10. Click **Finish**.

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